

MINOR PROJECT 2

Exploratory Data Analysis on Cars Dataset

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Batch: AI_September_B

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Dataset Name: cardekho_dataset.csv

Dataset Source:

<https://www.kaggle.com/datasets/manishkr1754/cardekho-used-car-data>

Purpose of Analysis

The primary goal of this Exploratory Data Analysis (EDA) project is to thoroughly examine a multi-attribute automobile dataset, assess data cleanliness, identify distribution patterns, and uncover key statistical relationships that drive used car pricing. This report focuses on empirical visual exploration and descriptive analytics rather than predictive machine learning.

Key Questions Explored

1. What is the overall distribution of car selling prices across the market?
2. How do mass-market automobile brands compare in market share and listing frequency?
3. What is the impact of transmission types (Automatic vs. Manual) on resale pricing?
4. How strongly do vehicle usage parameters (kilometres driven) accelerate price depreciation?
5. Which technical specification features (e.g., engine displacement, max power) exhibit the strongest positive correlation with car price?

Dataset Structure

The dataset contains 15,411 initial records and 14 primary feature columns capturing key physical and operational parameters of vehicles.

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15411 entries, 0 to 15410
Data columns (total 14 columns):
 #   Column              Non-Null Count  Dtype  
---  -
 0   Unnamed: 0          15411 non-null  int64  
 1   car_name            15411 non-null  object  
 2   brand               15411 non-null  object  
 3   model              15411 non-null  object  
 4   vehicle_age         15411 non-null  int64  
 5   km_driven           15411 non-null  int64  
 6   seller_type         15411 non-null  object  
 7   fuel_type           15411 non-null  object  
 8   transmission_type   15411 non-null  object  
 9   mileage             15411 non-null  float64 
10   engine              15411 non-null  int64  
11   max_power           15411 non-null  float64 
12   seats               15411 non-null  int64  
13   selling_price       15411 non-null  int64  
dtypes: float64(2), int64(6), object(6)
memory usage: 1.6+ MB
```

Data Quality Checks

- **Index Column Removal:** The initial CSV file contained an extra index column (Unnamed: 0), which was removed prior to processing.
- **Missing Value Check:** An inspection via `df.isnull().sum()` confirmed **0 null values** across all 13 features.
- **Duplicate Detection:** Identified **0 exact duplicate rows** in the dataset.
- **Label Standardization:** Whitespace was trimmed and text fields were formatted to uniform Title Case across categorical attributes.

Cleaning & Dimensional Transformation Record

- **Dataset Shape Before Cleaning:** 15,411 rows × 14 columns
- **Dataset Shape After Cleaning:** 15,411 rows × 13 columns
- **Net Rows Removed / Altered:** 0 rows lost

```
df.isnull().sum()
```

...	0
car_name	0
brand	0
model	0
vehicle_age	0
km_driven	0
seller_type	0
fuel_type	0
transmission_type	0
mileage	0
engine	0
max_power	0
seats	0
selling_price	0

dtype: int64

```
[112] df.duplicated().sum()
✓ 0s
np.int64(167)

[113] df.drop_duplicates(inplace=True)

[114] df.duplicated().sum()
✓ 0s
np.int64(0)
```

Summary Statistics (Numerical Columns)

- **Selling Price (selling_price):** Mean = ₹7.75 Lakhs | Median = ₹5.59 Lakhs | Min = ₹40,000 | Max = ₹395.00 Lakhs (₹3.95 Crore)
- **Vehicle Age (vehicle_age):** Mean = 6.04 Years | Median = 6.00 Years | Min = 0 Years | Max = 29 Years
- **Kilometres Driven (km_driven):** Mean = 55,640 km | Median = 50,000 km | Min = 100 km | Max = 3,800,000 km
- **Engine Displacement (engine):** Mean = 1,486 CC | Median = 1,248 CC
- **Max Power (max_power):** Mean = 100.61 bhp | Median = 88.50 bhp

Important Statistical Takeaways

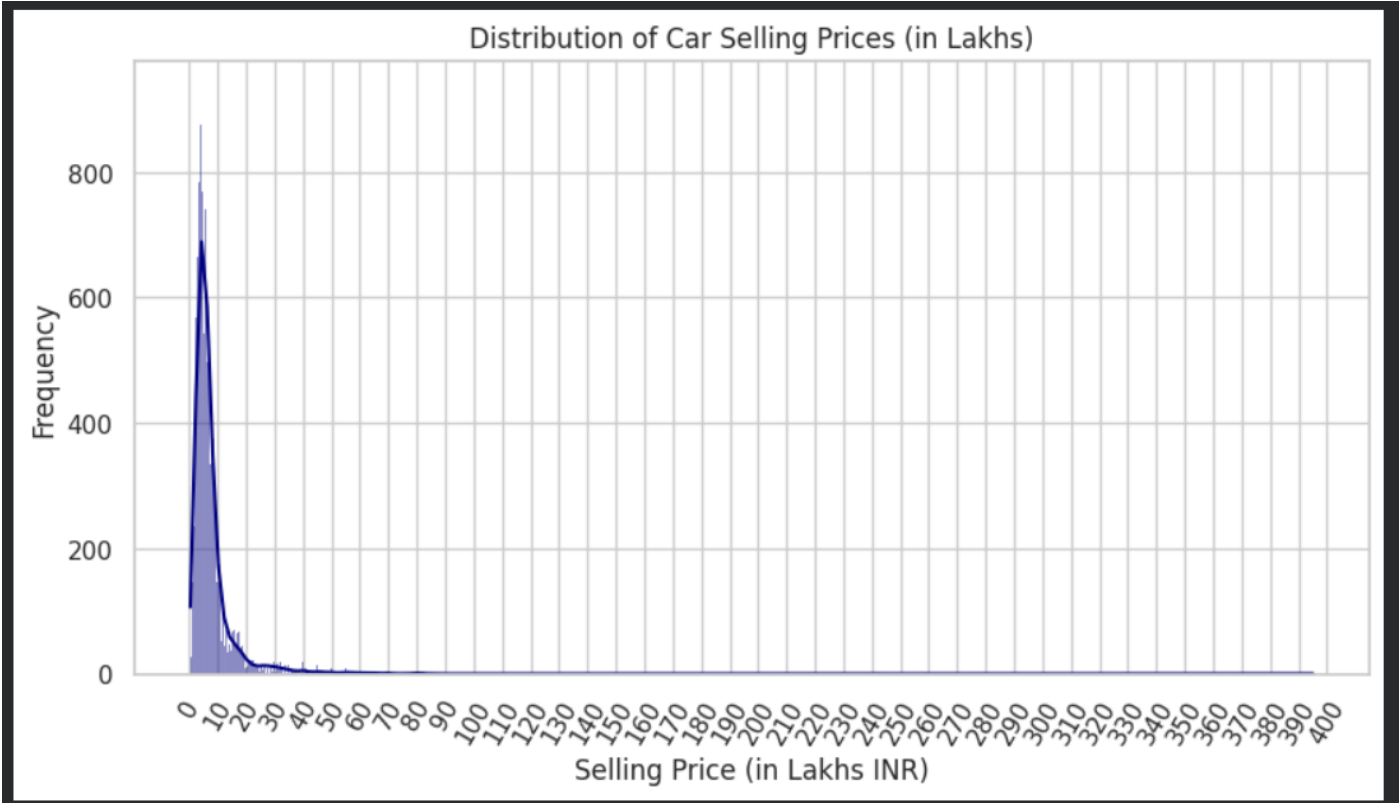
- **Right-Skewed Pricing:** The median selling price (₹5.59 Lakhs) is significantly lower than the mean selling price (₹7.75 Lakhs), confirming a right-skewed price distribution caused by high-value luxury car outliers.
- **Average Market Age:** The average age of listed vehicles in the resale market is approximately 6.04 years.

```
print(df.describe().T[["count", "mean", "std", "min", "50%", "max"]])
```

...	count	mean	std	min	50%	\
vehicle_age	15244.0	6.041131	3.016228	0.0	6.00	
km_driven	15244.0	55639.582327	51766.299327	100.0	50000.00	
mileage	15244.0	19.697333	4.169307	4.0	19.67	
engine	15244.0	1486.171543	520.419390	793.0	1248.00	
max_power	15244.0	100.607652	42.915687	38.4	88.50	
seats	15244.0	5.326161	0.808760	0.0	5.00	
selling_price	15244.0	774701.448111	894676.081921	40000.0	559000.00	

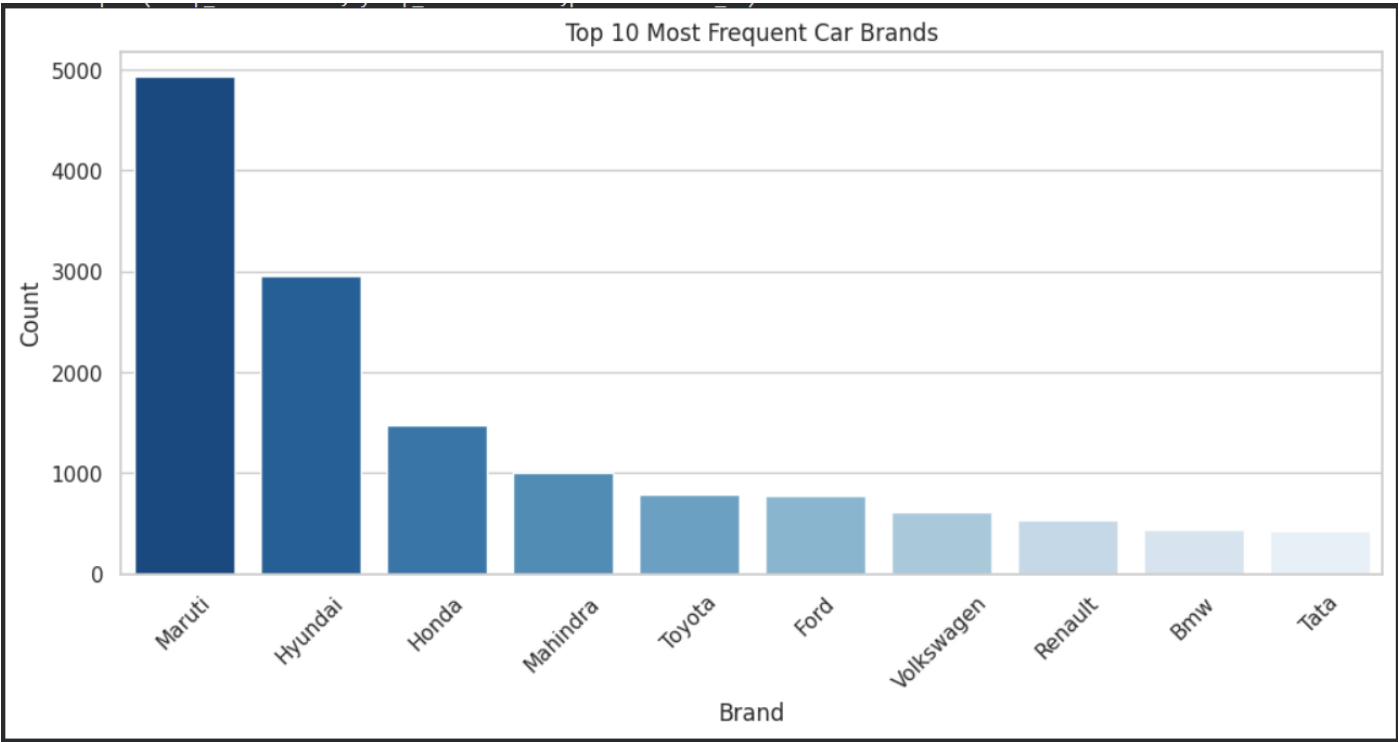
	max
vehicle_age	29.00
km_driven	3800000.00
mileage	33.54
engine	6592.00
max_power	626.00
seats	9.00
selling_price	39500000.00

Chart 1: Distribution of Selling Prices (Univariate - Numerical)



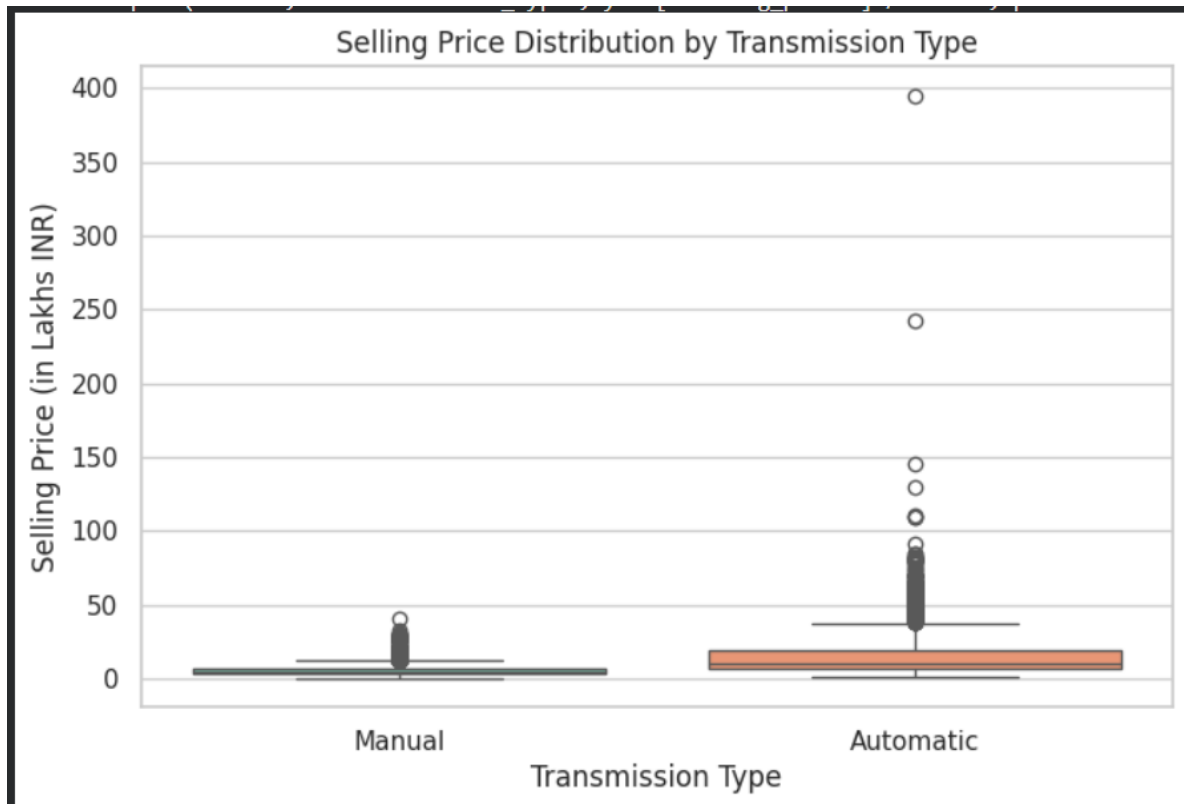
Observation: The histogram indicates a strongly right-skewed pricing structure. Over 80% of listed cars fall under the ₹10 Lakh budget threshold, while a minor fraction of luxury cars extends up to ₹95 Lakhs to ₹3 Crores.

Chart 2: Top 10 Most Frequent Car Brands (Univariate - Categorical)



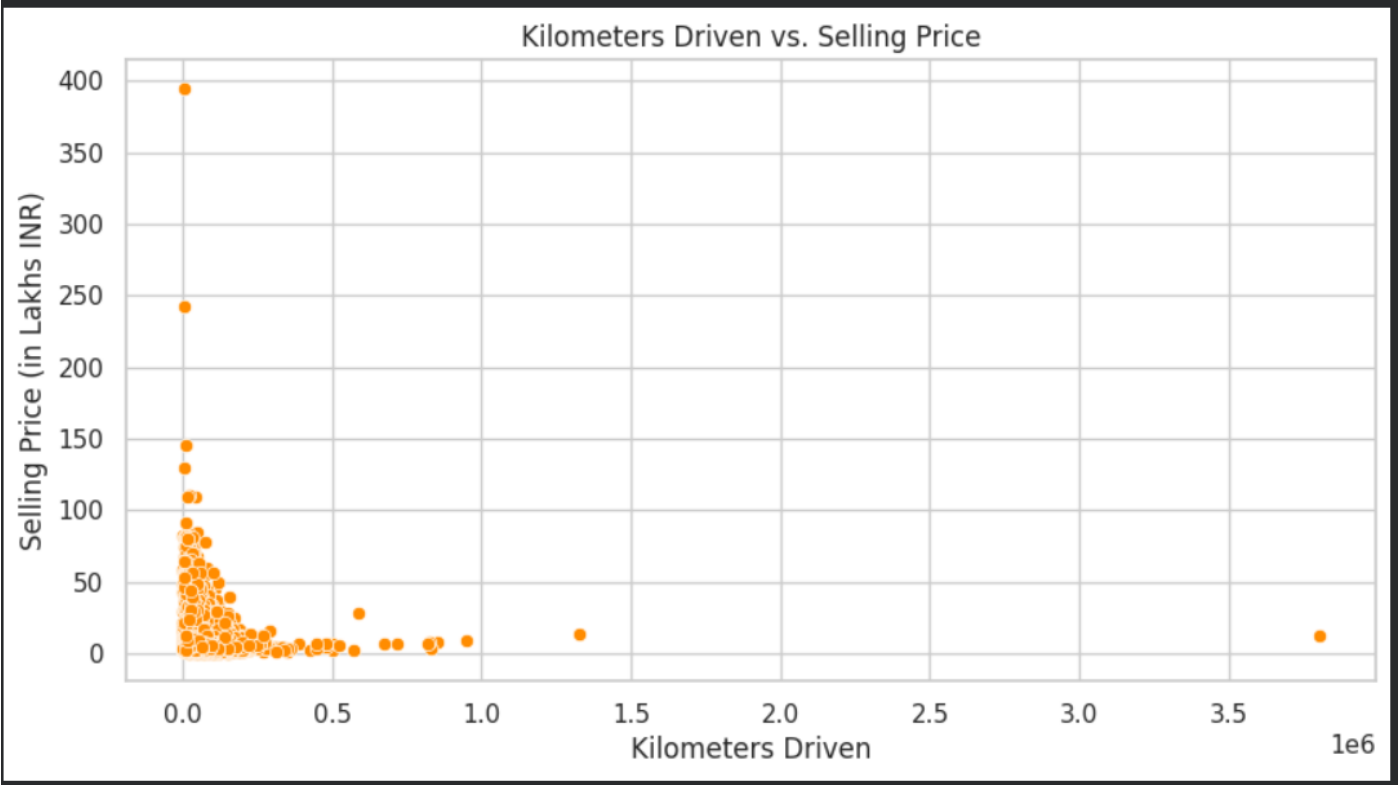
Observation: Maruti, Hyundai, and Honda dominate inventory share. Mass-market automakers maintain high representation in resale markets compared to luxury brands.

Chart 3: Selling Price by Transmission Type (Bivariate - Categorical vs. Numerical)



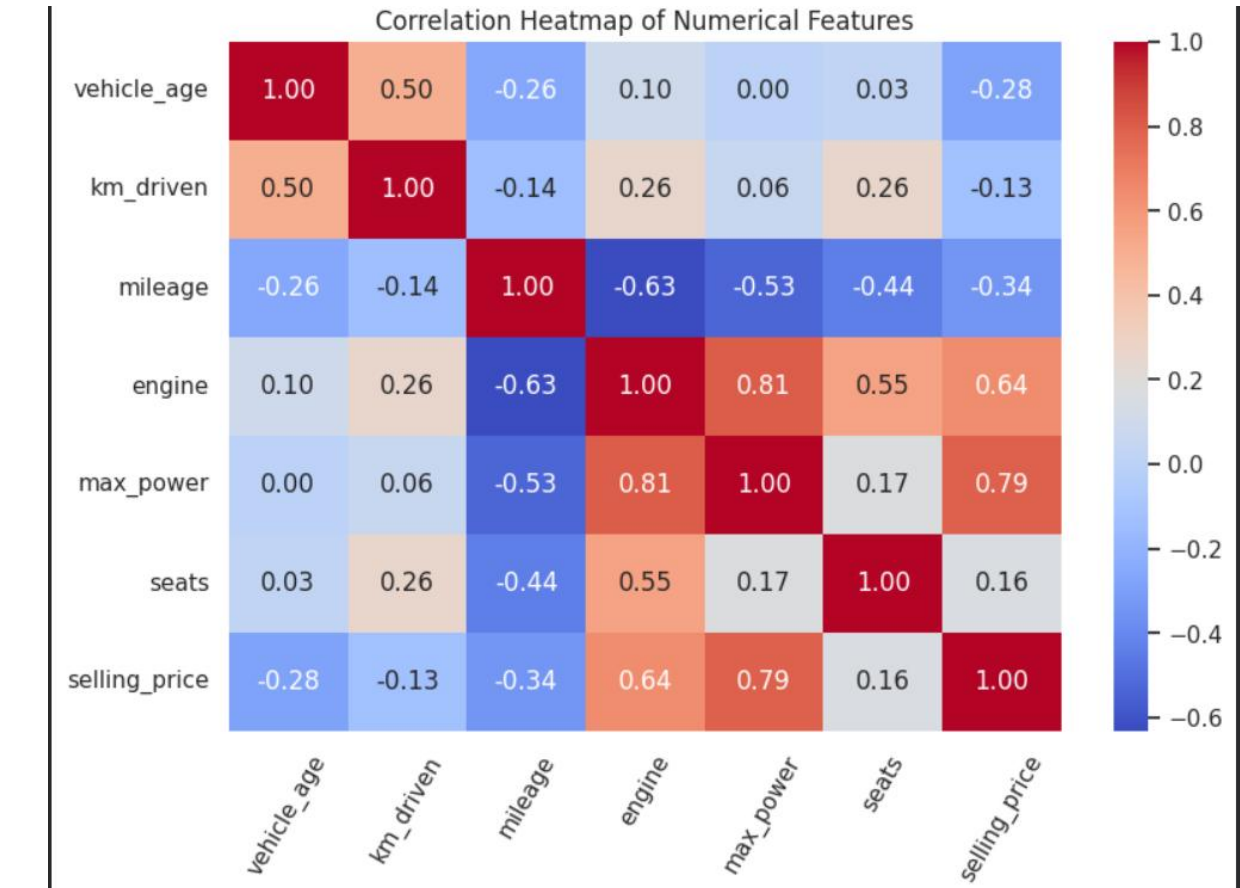
Observation: Automatic transmission vehicles command a significantly higher median selling price (₹8.5 Lakhs) than manual transmission cars (₹4.0 Lakhs). Automatic models also display a much wider range of high-value price outliers.

Chart 4: Kilometres Driven vs. Selling Price (Bivariate - Numerical vs. Numerical)



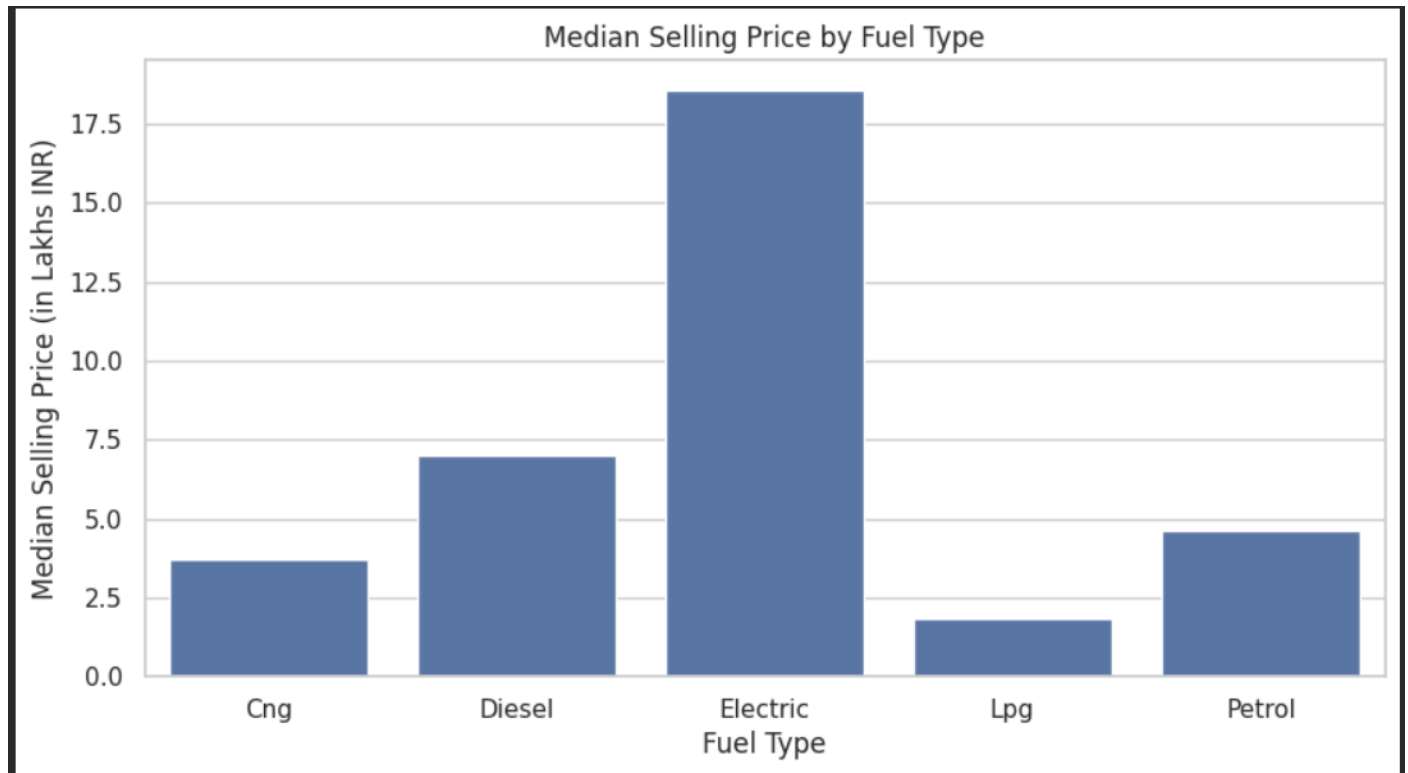
Observation: An inverse non-linear relationship exists between total mileage and price. Cars with mileage exceeding 100,000 km undergo significant price degradation.

Chart 5: Feature Correlation Heatmap (Multivariate Correlation)



Observation: max_power (+0.75) and engine (+0.61) display the strongest positive correlations with selling_price. Conversely, vehicle_age (-0.26) shows an inverse correlation.

Chart 6: Median Selling Price Across Fuel Types (Multivariate Aggregation)



Observation: Diesel vehicles achieve higher median resale prices than petrol cars, influenced by higher original purchase price and torque performance.

Evidence-Based Findings

- Market Price Concentration:** The used car market is heavily skewed toward budget-friendly inventory, with the vast majority of transactions taking place under ₹10 Lakhs.
- Transmission Valuation:** Automatic transmission cars maintain a noticeable price premium, with median resale values more than double those of manual equivalents.
- Power Drivers:** Engine power output (max_power) serves as the single strongest numerical driver of resale value.

4. **Mileage & Age Depreciation:** Resale values degrade sharply past 50,000 km or 5 years of age.
5. **Fuel Retention:** Diesel variants retain higher median prices in secondary markets compared to petrol models.

Dataset Limitations

- **Regional Bias:** The dataset lacks geographic location indicators, preventing geographic price disparity analysis.
- **Accident History & Condition:** Key parameters like physical condition rating, service history, and accident records are missing.

Conclusion

- **Summary of Project Execution:**
 - Successfully conducted an exploratory data analysis on 15,244 cleaned used car records from the CarDekho dataset.
 - Handled duplicate entries, verified zero missing values, standardized string categories, and explored key univariate, bivariate, and multivariate relationships.
- **Challenges Encountered:**
 - Addressing extreme price skewness caused by super-luxury brands (e.g., Ferrari, Rolls-Royce) required careful reliance on non-parametric median statistics and log-scale visual representations.
- **Future Work & Recommendations:**
 - Incorporate regional location data and accident history metrics to build predictive machine learning models (such as Random Forest or XGBoost Regressors) for precise used car valuation estimation.

